Mapping Inequality on the Boston Plate

Neighbourhood Socioeconomics & Food Inspection Outcomes in Boston, 2015–2025

Class of 2026 · **Juliette Bacuvier** · Capstone Project

RESEARCH QUESTION

Do food safety inspection outcomes in Boston systematically differ across neighbourhoods of varying socioeconomic status — and has any such gap widened over the past decade?

MAIN RESULT

Yes, but the effect is small — and it has not widened. Establishments in low-income tracts have a ∼3.4 percentage-point lower predicted pass probability than those in high-income tracts, and this gap has remained statistically stable across the 2015–2025 window. Most variation in pass/fail is within-establishment noise rather than neighbourhood-level disparity.

i. Background & Motivation

Public food inspection records document day-to-day enforcement of health regulations across an entire city. If neighbourhood disadvantage shapes whether establishments pass or fail, it would suggest disparities in regulatory protection across communities. Boston offers a ten-year, geocoded inspection record linked to census tract demographics, giving us a natural test case for asking whether neighbourhood SES predicts inspection outcomes after accounting for establishment-level differences.

ii. Data Sources

Ten years of Boston inspections linked to tract-level Census demographics.

Boston Food Establishment Inspections (2015–2025) — 872,111 raw violation records, deduplicated to 100,211 unique inspections at the establishment-visit level. **ACS 2019 5-year estimates** at census-tract level for income, poverty, racial composition, and foreign-born share. Spatially joined to inspections via tract polygons.

The overall 46% fail rate reflects Boston's inclusive definition of *fail* — any inspection with at least one violation of any severity. Most failed inspections involve minor, correctable issues rather than public health emergencies.

91.6K	6,912	174	46%
INSPECTIONS	ESTABLISHMENTS	CENSUS TRACTS	OVERALL FAIL RATE

iii. Methods

- **Mixed-effects logistic regression** via `glmmTMB` with random intercepts by establishment.
- **Sequential model building**: null → income-only → SES composite → fully adjusted → income × time interaction.
- **PCA-based SES index** to address collinearity ($r = -0.81$ income–poverty). PC1 captures 79%.
- **Train/test split** by establishment (80/20) to avoid leakage.
- **Robustness**: COVID exclusion, nonwhite check, DHARMa diagnostics.
- **Model diagnostics**: residual analysis via DHARMa, AUC evaluation on held-out establishments.

★ KEY FINDING

*A 1 SD increase in tract income (∼$40K) raises the odds of passing by **6.4%**. A low-income Food Service establishment has a **50.1%** predicted pass probability, vs **53.5%** in a high-income tract. The gap is real but small.*

iv. Exploratory Findings

At the tract level, the bivariate relationship between fail rates and median income is **weakly negative but visually flat**, which implies that any neighbourhood SES effect will be modest.

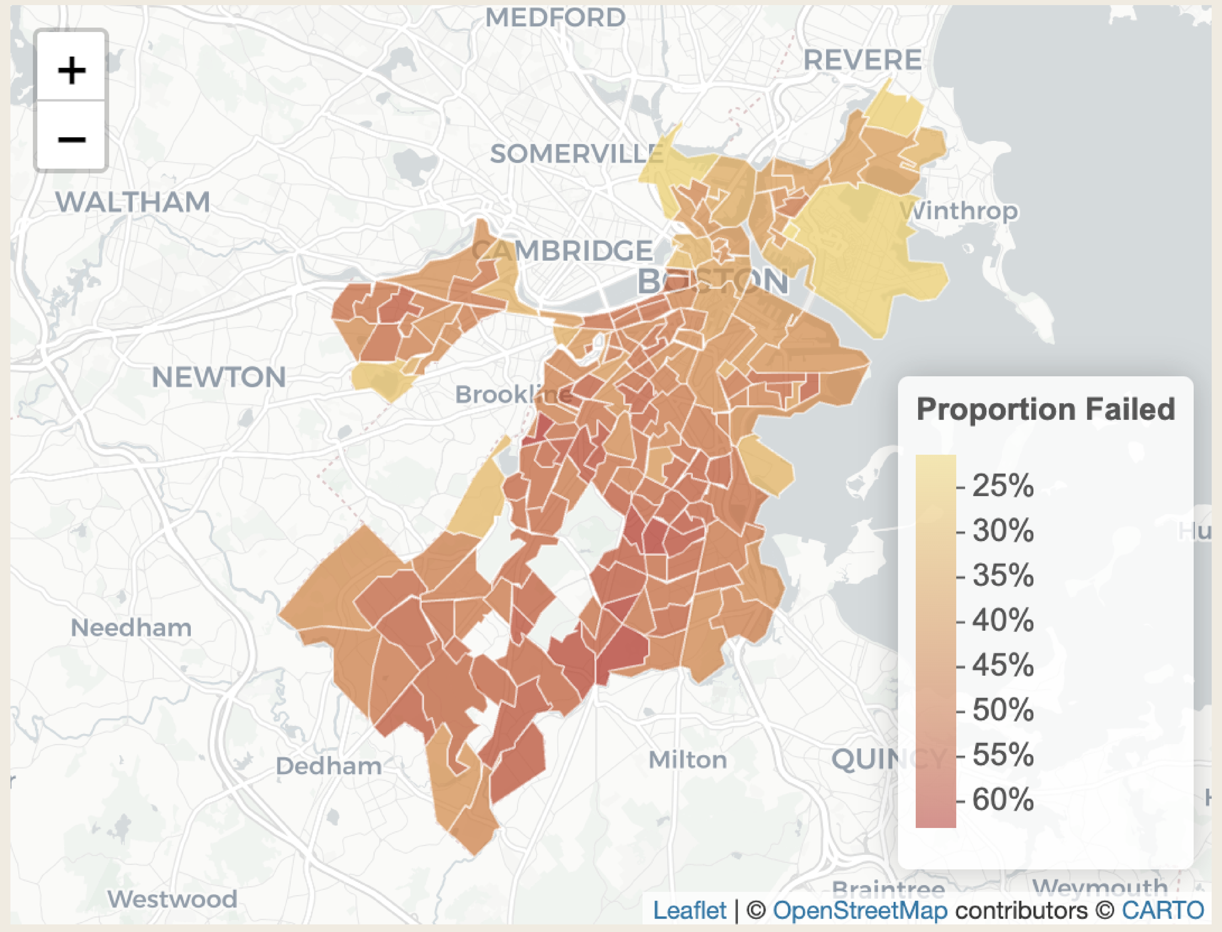

Figure 1. *Inspection fail rates by Boston census tract. Darker tracts indicate higher proportions of failed inspections; the spatial pattern motivates the neighbourhood-level analysis.*

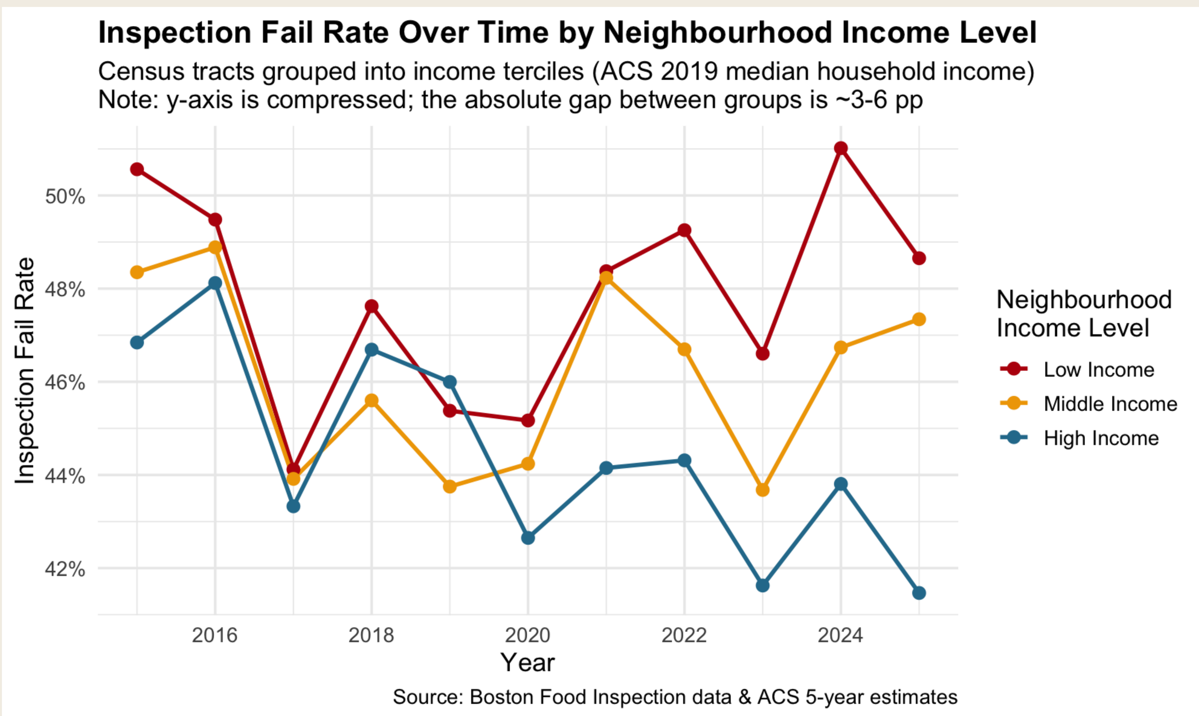

Figure 2. *Tract-level fail rates show only a weak negative slope across the income distribution. Each bubble is a census tract sized by inspection volume.*

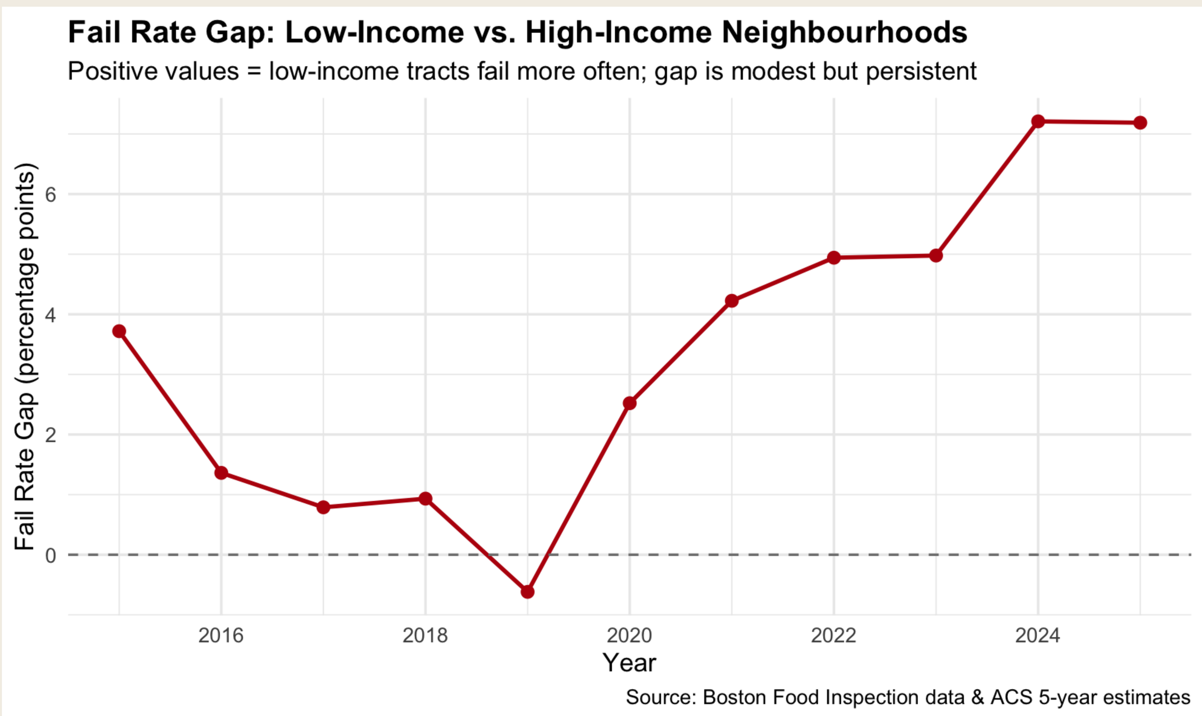

Figure 3. *Low-income tracts consistently fail more often than high-income ones, but the absolute gap is only* ∼*3–6 pp on a compressed y-axis.*

v. Modelling Results

The **null model ICC is just 0.046** — only 4.6% of pass/fail variation lies between establishments. The remaining 95% is within-establishment variation from visit to visit, suggesting that inspection outcomes are dominated by visit-level factors (inspector, day, conditions) rather than stable establishment characteristics. This places a hard ceiling on how much any neighbourhood-level predictor can explain. Adding fixed effects further shrinks the random-effect variance by **87.7%**, leaving very little residual establishment heterogeneity.

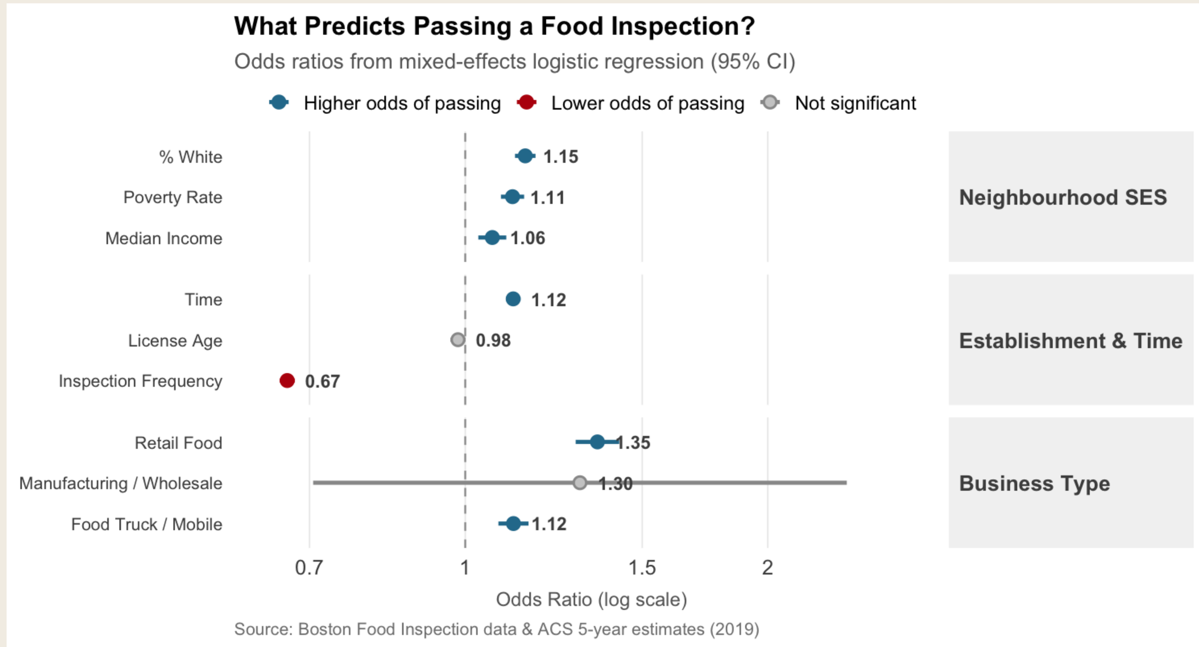

Figure 4. *Inspection frequency dominates the coefficient plot (OR ≈ 0.67). Median income is positive but modest (OR ≈ 1.06).*

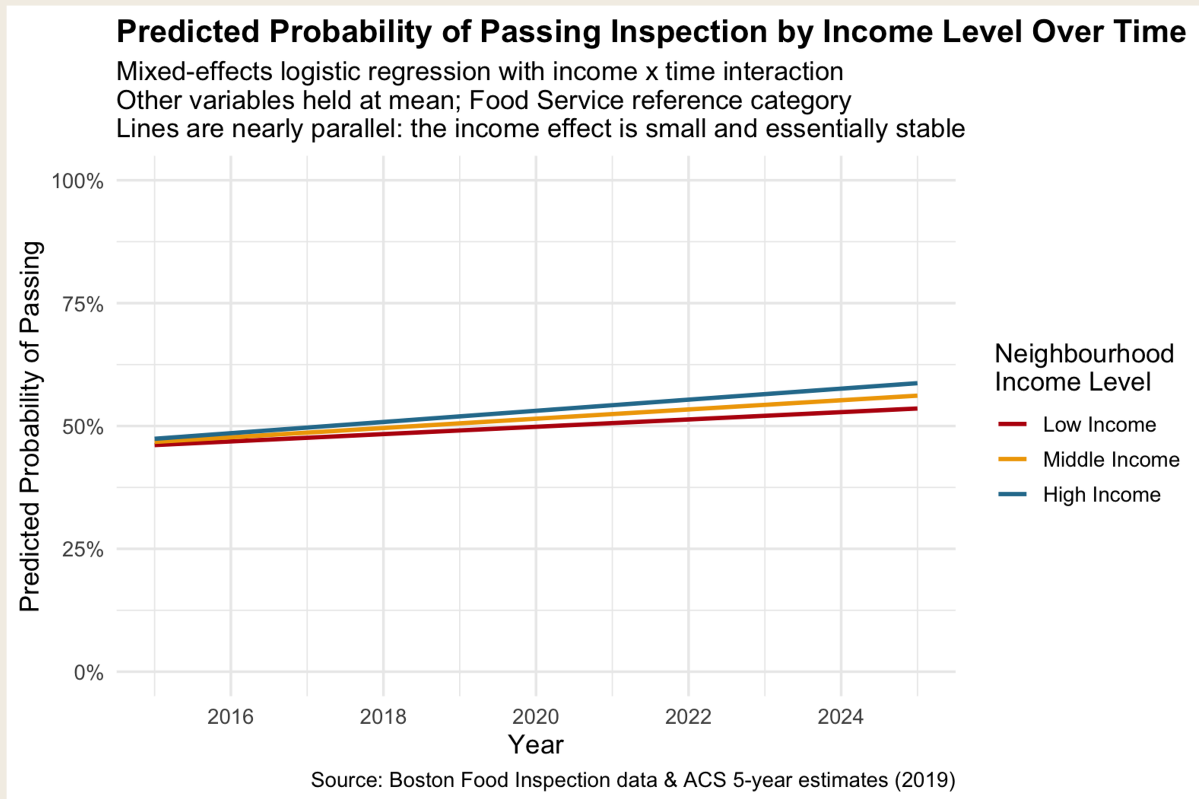

Figure 5. *Predicted-probability lines for low/middle/high income tracts are nearly parallel — the income × time interaction is significant ($p = 0.006$) but practically negligible ($OR = 1.022$).*

Limitations

Single-city analysis. ACS 2019 demographics applied to a 10-year window. Pass/fail is coarse and may not capture violation severity. Inspection frequency is endogenous to outcomes. Spatial autocorrelation not modelled. COVID sensitivity check showed all ORs shift < 0.02; $AUC = 0.624$.

vi. Interpretation Caveats

- **Poverty coefficient flips sign** when conditioning on income and % White (unadjusted $OR = 0.98 \rightarrow$ adjusted $OR = 1.11$) — a classic suppression effect from collinearity, *not* evidence that poverty improves outcomes. The PCA SES index ($OR = 1.065$) is the cleanest single measure.
- **Inspection frequency is endogenous** to outcomes: failures trigger re-inspections, so conditioning on frequency risks collider bias on the SES coefficient. A sensitivity check excluding frequency shifted the income OR by less than 0.01, suggesting the bias is small in practice — but this remains the model's most consequential limitation.
- **Business type explains more than SES** (Mobile food OR ≈ 1.42; Retail food OR ≈ 1.18), reinforcing that *what* an establishment does matters more than *where* it sits.

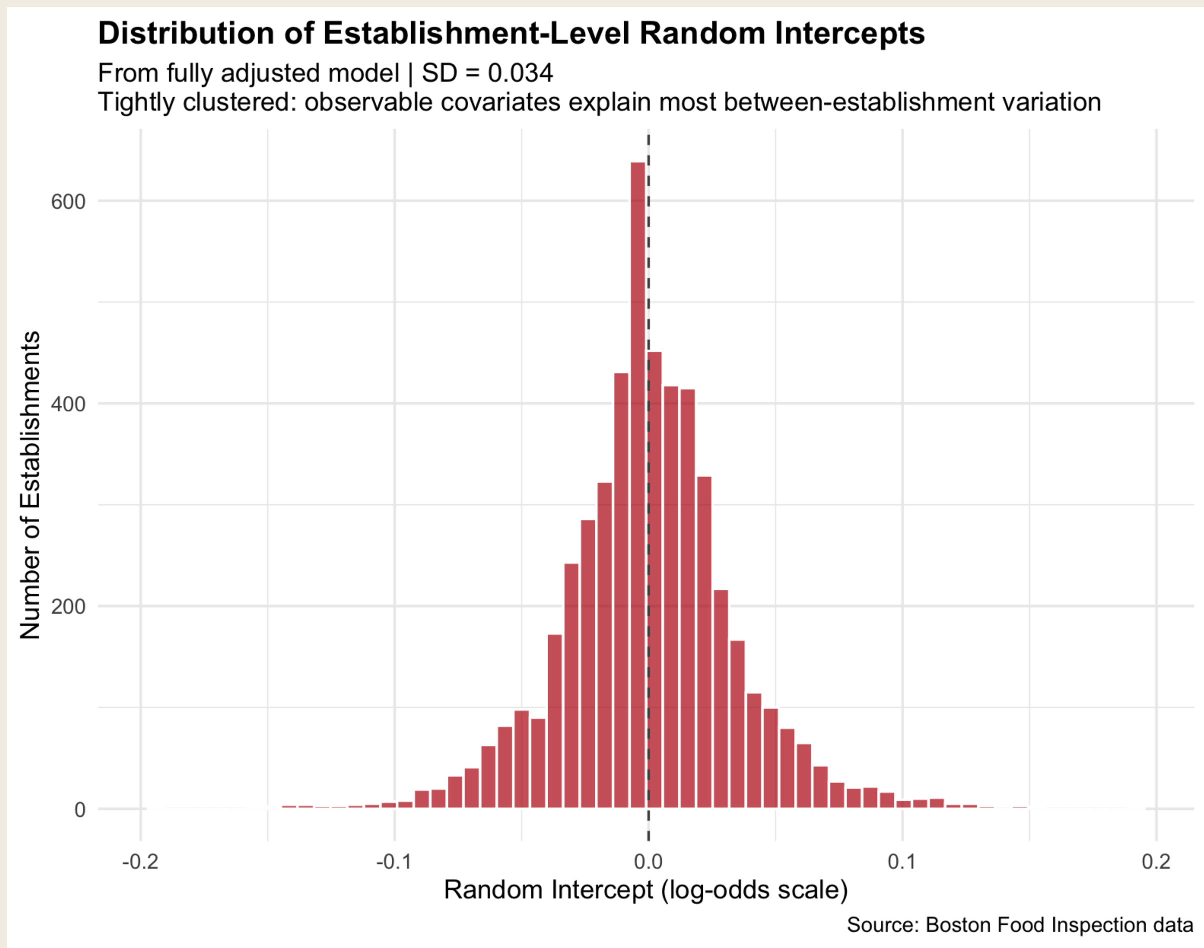

Figure 6. *Establishment random intercepts cluster tightly around zero ($SD = 0.034$) once observable covariates are accounted for — confirming little residual heterogeneity.*

vii. Conclusion

Neighbourhood income has a real but modest association with food inspection outcomes in Boston.

The effect is dwarfed by within-establishment variability and by differences across business types. The income gradient is statistically stable over the past decade.

These findings are consistent with a case for **establishment-level rather than area-based regulatory attention**, and push back against narratives that low-income Boston neighbourhoods systematically fail inspections at dramatically higher rates than wealthier ones. Whether establishment-level intervention would actually *improve* outcomes is a separate question this analysis cannot answer.